How BARCS Finds Important Genes

Chapter 2: The calculations

Prepared for Hyun-Hwan Jeong and Kyu-Won Lee

Companion slides to the student guide · BARCS 0.2.1 · Grades 7 to 9

The plan for Chapter 2

6. How BARCS finds the best β values
7. How BARCS measures the biological noise
8. Is the effect real?
9. Borrow information from other guides
10. Check the results with control guides
11. From guides to genes
12. Many genes and false discoveries

Remember

Each section has the same title as a step in Chapter 2 of the textbook. The numbers are real BARCS 0.2.1 output for our example.

What you need from Chapter 1

Our example and the key ideas

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

- Proportions: 191 (control) and 111 (drug) per million. Fold change 0.58.
- Log-odds: -8.56 (control), -9.10 (drug).
- Variance: $N\mu(1-\mu)\big(1+(N-1)\rho\big)$.
- Design matrix X rows: $(1,0)$, $(1,0)$, $(1,1)$, $(1,1)$.
- Prediction η = row of X times β; $\mu = e^{\eta}/(1+e^{\eta})$.
- β_0: control log-odds. β_1: effect of the drug.

Step 6: How BARCS finds the best β values

Guess, check, improve

BARCS uses a loop: iteratively reweighted least squares (IRLS).

1. **Guess.** Start with the simple log-odds of each group.
2. **Predict.** Use X to predict η and μ for each library.
3. **Measure the noise.** Find ρ (Step 7).
4. **Weigh.** Less noise gives a larger weight.
5. **Correct.** Calculate a working value for each library.
6. **Improve.** Weighted averages give the new β values.
7. **Repeat** until the β values stop changing.

The formulas

Weight of library i:

$$w_i = \frac{N_i \, \mu_i \, (1 - \mu_i)}{1 + (N_i - 1) \, \rho}$$

Working value of library i (prediction plus a correction):

$$v_i = \eta_i + \frac{K_i / N_i - \mu_i}{\mu_i \, (1 - \mu_i)}$$

For our design matrix, the new β values are weighted averages:

$$\beta_0 = \frac{w_1 v_1 + w_2 v_2}{w_1 + w_2}, \qquad \beta_0 + \beta_1 = \frac{w_3 v_3 + w_4 v_4}{w_3 + w_4}$$

Iteration 1: guess, predict, measure the noise

Worked example 10a: iteration 1

Guess. Control log-odds -8.563522, drug log-odds -9.104869.
So $\beta_0 = -8.563522$ and $\beta_1 = -9.104869 - (-8.563522) = -0.541347$.

Predict. Control $\mu = 0.000190909$. Drug $\mu = 0.000111111$.

Measure the noise. Step 7 gives $\rho = 0.0000003841$.

Iteration 1: weigh and correct library 1

Worked example 10b: library 1

Weigh.

Top: $1{,}000{,}000 \times 0.000190909 \times 0.999809 = 190.87$

Bottom: $1 + 999{,}999 \times 0.0000003841 = 1.3841$

Weight: $w_1 = 190.87 \div 1.3841 = 137.90$

Correct.

$K_1/N_1 - \mu_1 = 0.000180 - 0.000190909 = -0.0000109$

$-0.0000109 \div 0.000190873 = -0.0572$

Working value: $v_1 = -8.5635 + (-0.0572) = -8.6207$

Iteration 1: improve

Lib.	w_i	v_i	$w_i \times v_i$
1	137.905	−8.62068	−1188.84
2	156.785	−8.51589	−1335.16
3	80.269	−9.02486	−724.42
4	67.988	−9.20488	−625.82

Worked example 10c: new values

Control:

$(-1188.84 - 1335.16) \div 294.69 = -8.564929$

Drug:

$(-724.42 - 625.82) \div 148.257 = -9.107414$

$\beta_1 = -9.107414 - (-8.564929) = -0.542486$

The next iterations

After iteration	β_0	β_1
0 (the guess)	-8.563522	-0.541347
1	-8.564929	-0.542486
2	-8.564934	-0.542492
3	-8.564934	-0.542492

After iteration 3 nothing changes. Final: $\beta_0 = -8.565$, $\beta_1 = -0.542$.

Remember

The weights move β_1 from -0.5413 to -0.5425. Library 2 has the most reads and the least relative noise, so it counts the most.

Step 7: How BARCS measures the biological noise

Pearson residuals and the rule for ρ

A Pearson residual is the distance from the prediction in standard deviations:

$$r_i = \frac{K_i - N_i\mu_i}{\sqrt{N_i\mu_i(1-\mu_i)(1+(N_i-1)\rho)}}$$

Remember

BARCS chooses ρ so that the squared Pearson residuals add up to the degrees of freedom:

$$r_1^2 + r_2^2 + r_3^2 + r_4^2 = \text{df}$$

Start with $\rho = 0$

Worked example 11

Library	$(K - N\mu)^2$	$N\mu(1-\mu)$	Squared residual
1	$(-10.64)^2 = 113.21$	190.60	0.594
2	$(+11.23)^2 = 126.16$	228.72	0.552
3	$(+9.17)^2 = 84.14$	110.82	0.759
4	$(-8.66)^2 = 75.03$	88.65	0.846
		Sum	2.751

The sum (2.751) is bigger than df $= 2$. So there is more noise than counting noise: $\rho > 0$.

Guess and check for ρ

ρ	Sum
0	2.751
0.0000001	2.506
0.0000002	2.302
0.0000003	2.129
0.000000386	2.000
0.0000005	1.851

- A larger ρ makes the sum smaller.
- BARCS stops when the sum equals 2: $\rho = 0.000000386$.
- If the sum is 2 or less at $\rho = 0$, BARCS uses $\rho = 0$.

Variance inflation

$2.751 \div 2 = 1.376$: the guide has 1.38 times more noise than counting noise alone (used in Step 9).

Step 8: Is the effect real?

The standard error

The standard error (SE) tells how much β_1 can change because of noise. For our design matrix, it comes from the weights:

$$W_{\text{control}} = w_1 + w_2, \quad W_{\text{drug}} = w_3 + w_4, \qquad \text{SE}(\beta_1) = \sqrt{\frac{1}{W_{\text{control}}} + \frac{1}{W_{\text{drug}}}}$$

Worked example 13

Final weights: 137.524, 156.322, 79.955, 67.737.

$W_{\text{control}} = 293.846, \quad W_{\text{drug}} = 147.692$

$1/293.846 + 1/147.692 = 0.003403 + 0.006771 = 0.010174$

$\text{SE} = \sqrt{0.010174} = 0.101$

The t value and the degrees of freedom

$$t = \frac{\beta_1}{\text{SE}}, \qquad \text{df} = (\text{rows of } X) - (\text{columns of } X)$$

- A t value far from 0: the effect is large compared with the noise.
- Fewer degrees of freedom: BARCS needs a bigger t.

Degrees of freedom	1	2	14	very many
$\lvert t \rvert$ needed for $p < 0.05$	12.71	4.30	2.14	1.96

The p-value

- The p-value: if the true effect is 0, how often does noise alone give a t this far from 0?
- Small p-value: noise alone seldom gives such a result. The usual limit is 0.05.

Worked example 14: testing the effect

1. $t = -0.542 \div 0.101 = -5.38$
2. X has 4 rows and 2 columns: $\text{df} = 2$
3. $\text{df} = 2$ needs $|t| > 4.30$. Our $|t| = 5.38$. So $p < 0.05$.
4. Exact p-value: $p = 0.033$

Degrees of freedom come from libraries

Remember

The degrees of freedom come from the number of libraries, not from the number of reads.

A library with 100 million reads still counts as one row of X.

Step 9: Borrow information from other guides

Moderation: a weighted vote

- 2 degrees of freedom give an unreliable noise estimate.
- A screen has thousands of guides with similar noise.
- BARCS averages the guide's own inflation with the typical inflation.

$$\text{moderated} = \frac{\text{df} \times \text{own} + d_0 \times \text{typical}}{\text{df} + d_0}$$

What the symbols mean

d_0: the prior degrees of freedom, the votes of the typical value. BARCS calculates it from all the guides.

Worked example: moderation with simple numbers

Worked example 15

A guide has df $= 3$ and own inflation 4.0. Typical inflation 1.5 with $d_0 = 12$.

$$\frac{3 \times 4.0 + 12 \times 1.5}{3 + 12} = \frac{12 + 18}{15} = \frac{30}{15} = 2.0$$

The value moves from 4.0 toward 1.5 and stops at 2.0.

What moderation does to our guide

BARCS uses the moderated value to:

1. multiply the SE by $\sqrt{\text{moderated} \div \text{own}}$
2. recalculate t
3. use $\text{df} + d_0$ degrees of freedom

BARCS moderates when it has 50 or more usable guides.

Worked example 16

Own inflation 1.376, df $= 2$. Typical 1.5, $d_0 = 12$.

Moderated: $(2 \times 1.376 + 12 \times 1.5) \div 14 = 1.482$

SE factor: $\sqrt{1.482 \div 1.376} = 1.038$

New SE: $0.101 \times 1.038 = 0.105$

New t: $-0.542 \div 0.105 = -5.18$, df $= 14$

New p-value: 0.00014

Step 10: Check the results with control guides

Non-targeting control guides

- Control guides switch off no gene. Their true effect is 0.
- About 5% of them should get $p < 0.05$ by chance.
- More than that: the noise model is too small. BARCS makes the tests wider.

Worked example 17: the scale factor

100 control guides: we expect 5 with $p < 0.05$, but 15 have it.
95 of the 100 control guides have $|t|$ below 2.47. The table says 1.96.
Scale factor: $2.47 \div 1.96 = 1.26$. BARCS divides every t by 1.26: $t = 3.5$ becomes $3.5 \div 1.26 = 2.78$.

The scale factor is never less than 1. Calibration only makes results more careful.

Step 11: From guides to genes

The Stouffer method

1. Change each guide's p-value into a z score. The sign follows the effect.
2. Add the z scores.
3. Divide by the square root of the number of guides, m.

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m}}$$

Two-sided p-value	0.317	0.134	0.046	0.012	0.0027
z score	1.0	1.5	2.0	2.5	3.0

Worked examples: combining guides

Worked example 18: three guides

p-values 0.046, 0.134, 0.012 give $z = 2.0$, 1.5, 2.5.

$$Z = \frac{2.0 + 1.5 + 2.5}{\sqrt{3}} = \frac{6.0}{1.732} = 3.46$$

$p = 0.00053$. Together, the guides give strong evidence.

Worked example 19: shared noise

Guides can share noise. The guide correlation r makes the bottom bigger: $\sqrt{m + m(m-1)r}$.

With $r = 0.1$: $\sqrt{3 + 6 \times 0.1} = \sqrt{3.6} = 1.897$.

$Z = 6.0 \div 1.897 = 3.16$, $p = 0.0016$.

Step 12: Many genes and false discoveries

Why we need a false discovery rate

- A screen tests about 20,000 genes.
- With $p < 0.05$, about 5% pass by chance: about 1,000 false results.
- The false discovery rate (FDR) is the expected fraction of false results in the list of hits.

Benjamini–Hochberg method:

1. Sort the genes by p-value and give them ranks.
2. Calculate $p \times$ (number of genes) $\div$ rank.
3. From the bottom up, replace a value with the value below it if that value is smaller.

Worked example: FDR for five genes

Worked example 20

Gene	Rank	p	$p \times 5 \div \text{rank}$	FDR
A	1	0.001	$0.001 \times 5 \div 1 = 0.005$	0.005
B	2	0.008	$0.008 \times 5 \div 2 = 0.020$	0.020
C	3	0.039	$0.039 \times 5 \div 3 = 0.065$	0.051 (from D)
D	4	0.041	$0.041 \times 5 \div 4 = 0.051$	0.051
E	5	0.600	$0.600 \times 5 \div 5 = 0.600$	0.600

FDR < 0.05: genes A and B. FDR < 0.10: genes A, B, C and D.

Two special cases

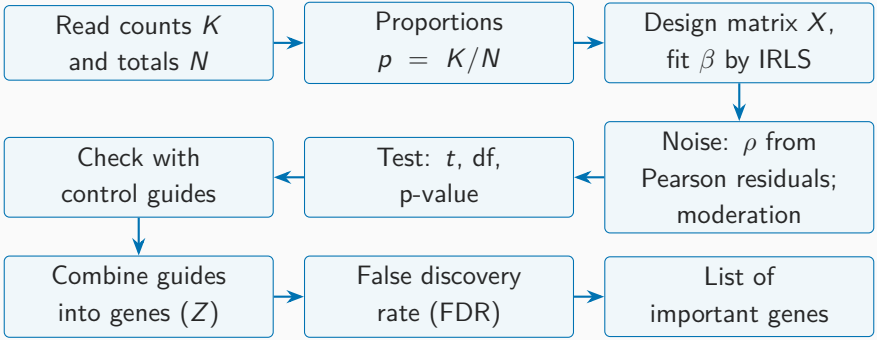
Zero reads and very large changes

A guide with zero reads
0 reads in all control libraries means log-odds $= \ln(0)$, which does not exist. Older BARCS removed the guide. BARCS 0.2.1 first adds less than one read (the Firth correction) and keeps it.

A very large change
A guide that grows hundreds of times can make the usual test fail: the SE grows faster than the effect. The likelihood-ratio test compares the fit with and without the effect. `test = "auto"` uses it only for such guides.

Summary: the full path

From counts to a list of genes



The symbols of Chapter 2

Symbol	Meaning	Value in our example
w	weight of a library in the fit	137.5 (library 1)
working value	prediction plus correction	-8.621 (library 1)
Pearson residual	distance from the prediction in SDs	-0.65 (library 1)
inflation	noise compared with counting noise	1.376
SE	standard error of the effect	0.101
t	effect divided by SE	-5.38
df	rows of X minus columns of X	$4 - 2 = 2$
p-value	chance of this result if the effect is 0	0.033
d_0	prior degrees of freedom (votes)	12
z, Z	z score of a guide, of a gene	3.46
m	number of guides for a gene	3
r	guide correlation (shared noise)	0.1
FDR	expected fraction of false hits	0.005 for gene A